

1. Describe the historical development of Building Technology from ancient civilizations to modern construction practices. Highlight the evolution of building technology in Nepal.
2. What is Building Technology? Explain its scope and discuss how advances in building technology have influenced the development of Nepalese buildings.
3. Explain the concept of occupancy classification of buildings. Why is occupancy classification important in building planning, design, and safety? Illustrate your answer with examples.
4. Explain the principles of building orientation and planning. Discuss how proper orientation contributes to energy efficiency, thermal comfort, natural lighting, and ventilation.
5. Define building planning and setting-out. Explain the principles of planning and describe the procedure of setting-out a building on the ground with neat sketches.
6. Discuss the importance of proper ventilation in buildings. Explain the principles of natural ventilation and describe the factors affecting its effectiveness.
7. What are acoustic defects? Explain different acoustic defects commonly found in auditoriums and lecture halls. Discuss suitable remedial measures for each defect.
8. Define a foundation and a basement. Explain the functions and requirements of a good foundation and discuss the advantages and disadvantages of providing a basement in a building.
9. What is site exploration? Explain its objectives and discuss the various methods of site exploration used before the design and construction of foundations.
10. What are deep foundations? Explain pile foundations and caisson foundations with neat sketches. Discuss their applications, advantages, and limitations.
11. Discuss the factors affecting the selection of a suitable foundation for a building. Explain how soil conditions, loading, groundwater level, and site constraints influence foundation choice.
12. Explain the methods of foundation excavation in soft soil, hard rock, and waterlogged (wet) conditions. Discuss the precautions and safety measures to be adopted during excavation.
13. What is basement waterproofing? Explain the causes of water leakage in basements and discuss the various waterproofing techniques used to prevent seepage.
14. Discuss different types of brick bonds (English, Flemish, Header, and Stretcher) with neat sketches, and compare their suitability for various construction purposes.
15. Describe the different types of stone masonry, namely Rubble Masonry and Ashlar Masonry. Explain their characteristics, advantages, disadvantages, and applications in modern construction.
16. Differentiate between load-bearing structures and framed structures. Explain their structural behavior, advantages, disadvantages, and suitability for residential and commercial buildings.

17. Explain the process of plastering and discuss the common defects in plastering. Also describe different types of pointing and their applications in masonry construction.
18. Discuss the methods of painting on wooden, metal, and masonry surfaces. Explain the surface preparation, painting procedure, and precautions required to ensure durable finishes.
19. Explain the different types of doors used in building construction. Describe the terminology of doors with neat sketches and discuss the suitability of each type for residential, commercial, and industrial buildings.
20. Discuss the different types of windows and ventilators used in buildings. Explain their terminology, functions, advantages, disadvantages, and selection criteria for different building requirements.
21. Discuss the different types of flooring used in building construction. Differentiate between solid floors and suspended floors, and explain various flooring materials, their construction methods, advantages, disadvantages, and applications.
22. Describe different electrical wiring systems used in buildings. Explain their installation methods, advantages, disadvantages, safety requirements, and suitability for residential and commercial structures.
23. Discuss the purpose, components, installation, and maintenance of lightning arresters in buildings. Explain their importance in protecting structures and occupants from lightning hazards.
24. Explain the different types of roofs used in building construction. Discuss flat, pitched, and curved roofs with neat sketches, highlighting their structural features, advantages, disadvantages, and suitable applications.
25. Describe timber roof construction. Explain the construction, components, advantages, disadvantages, and applications of King Post and Queen Post trusses with suitable sketches.
26. Explain the different types of roof covering materials used in buildings. Discuss the properties, advantages, disadvantages, installation methods, and applications of CGI sheets, tiles, and slates.
27. Explain the purpose of scaffolding in building construction. Describe the different types of scaffolding with neat sketches, and discuss their advantages, disadvantages, and suitable applications.
28. Define shoring and underpinning. Explain their objectives, types, construction methods, and applications in the repair, strengthening, and stabilization of existing structures.
29. Describe the causes of structural and non-structural cracks in buildings. Explain the methods of inspection, diagnosis, repair, and restrengthening techniques used to improve the safety and durability of damaged structures.
30. Discuss the concept of retrofitting in civil engineering. Explain the need for retrofitting and describe various retrofitting techniques used to enhance the strength, stability, and earthquake resistance of existing buildings.

31. Describe the major provisions of the National Building Code of Nepal (NBC). Explain how the code contributes to structural safety, public health, environmental protection, and quality construction practices.
32. Define green buildings and sustainable buildings. Explain their basic concepts, principles, objectives, and importance in promoting environmentally responsible and energy-efficient construction.
33. Explain the role of engineers in promoting compliance with the National Building Code of Nepal and the adoption of green building technologies. Discuss the professional responsibilities and ethical considerations involved in sustainable construction.